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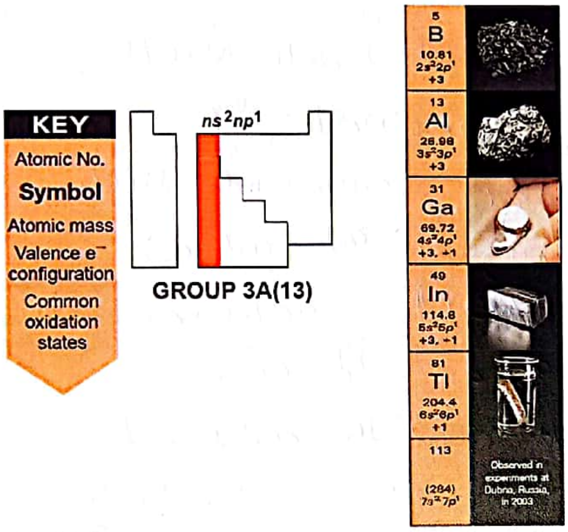
General Chemistry- Chemistry of Elements Day 04- 17/03/2022

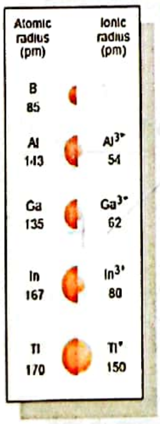
The Boron Family

14.5 GROUP 3A(13): THE BORON FAMILY

Group 3A(13) is the first of the p block.



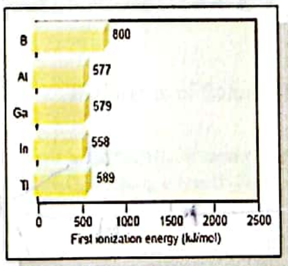
Atomic Properties

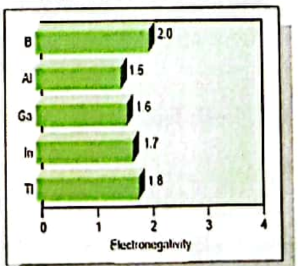


Group electron configuration is ns^2np^1. All except Tl (and presumably element 113) commonly display the +3 oxidation state. The +1 state becomes more common down the group.

Atomic size is smaller and EN is higher than for 2A(2) elements; IE is lower, however, because it is easier to remove an electron from the higher energy *p* sublevel.

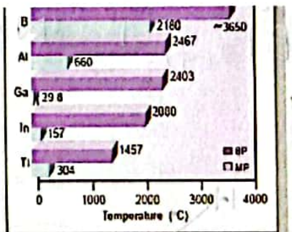
Atomic size, IE, and EN do not change as expected down the group because there are intervening transition and inner transition elements.

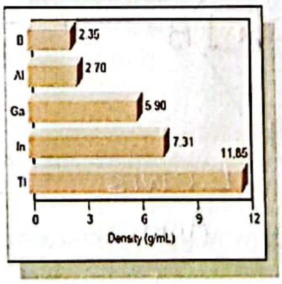




Physical Properties

Bonding changes from network covalent in B to metallic in the rest of the group. Thus, B has a much higher melting point than the others, but there is no overall trend. Boiling points decrease down the group.





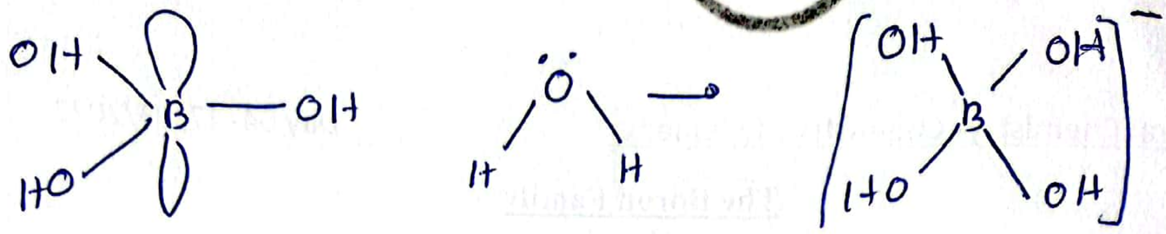
Densities increase down the group.

~~bot~~ BP↓ density↑

one ~~donatin~~ donating a pair of electrons and accepting by another to form a covalent bon

↓

Lewis acid-basic reaction.

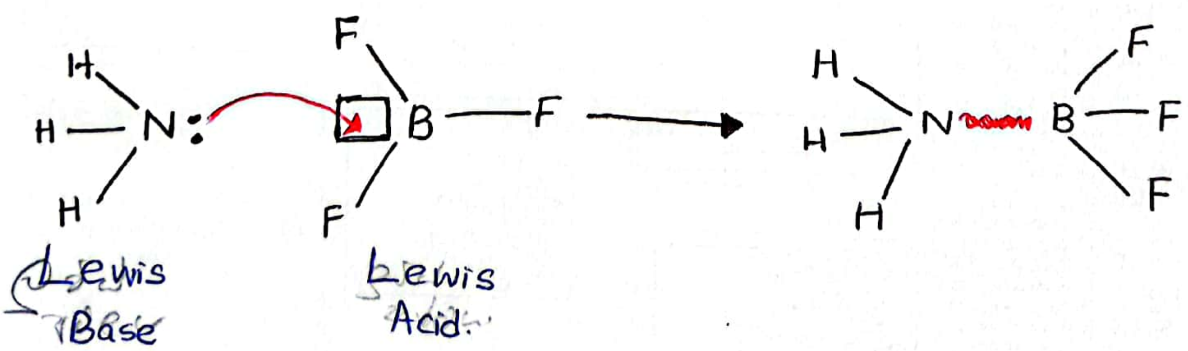


$H_3BO_3 + 3NaOH \rightarrow Na_3BO_3 + 3H_2O$

H_3BO_3 cannot be titrated satisfactory with NaOH as a sharp end point is not obtained. If certain organic polyhydroxic compounds such as glycerol, Mannitol or sugars are added to the mixture. Then $B(OH)_3$ behaves as a strong mono-basic acid. It can now titrated with NaOH and the end point is detected by Phynopthalin as in indicated.

$$B(OH)_3 + NaOH \rightleftharpoons Na[B(OH)_4] + NaBO_2 + 2H_2O$$

$$BF_{3(g)} + :NH_{3(g)} \longrightarrow F_3B - NH_{3(g)}$$



Lewis acid-base pairs.

The redox behavior of the elements in the group

1. multiple oxidation states present.
2. lower state becomes more prominent going down the group.
3. oxides with the element in a lower oxidation state are more basic than oxides with the element in a higher oxidation state.

Highlights of Boron Chemistry

Accepting a Bonding Pair from an Electron-Rich Atom

In gaseous boron trihalides (BX3), the B atom is electron deficient, with only six electrons around it. To attain an octet, the B atom accepts a lone pair of electrons from an electron-rich atom (red) and forms a covalent bond:

$$BF_3(g) + {:}NH_3(g) \longrightarrow F_3B\text{—}NH_3(g)$$

Such reactions, in which one reactant accepts an electron pair from another to form a covalent bond are known as Lewis acid-base reactions.

Highlights of Boron Chemistry

- Similarly, B has only six electrons in boric acid, $B(OH)_3$. In water, the acid itself does not release a proton. Rather, it accepts an electron pair from the O in H_2O, forming a fourth bond and releasing an H ion:

$$B(OH)_3(s) + H_2O(l) \rightleftharpoons B(OH)_4^-(aq) + H^+(aq)$$

- Boron also attains an octet in several boron-nitrogen compounds whose crystal or molecular structures are amazingly similar to those of elemental carbon and some of its organic compounds. These similarities are due to the fact that the size, IE, and EN of C are between those of B and N.

- Moreover, C has four valence electrons $\cdot\dot{\underset{\cdot}{C}}\text{—}\dot{\underset{\cdot}{C}}\cdot$ and $\cdot\dot{B}\text{—}\dot{\underset{\cdot}{N}}\cdot$ ı N has five, so the following grouping have the same number of valence electrons.

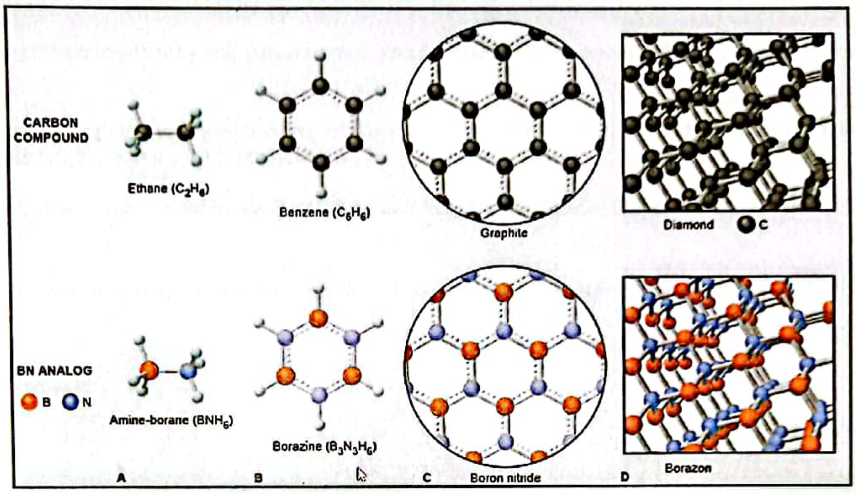
Therefore, $H_3C\text{-}CH_3$ (ethane) and $H_3B\text{-}NH_3$ (amine-borane) are isoelectronic.

$B_2O_3 > Al_2O_3 > Ga_2O_3 > In_2O_3 > Tl_2O$

weakly acidic $\xleftarrow{\text{Acidity}}$ strongly basic

Highlights of Boron Chemistry

- Boron nitride has a structure consisting of hexagons fused into sheets, very similar to that of graphite. Moreover, like graphite, its electrons are highly delocalized through resonance.



Highlights of Boron Chemistry

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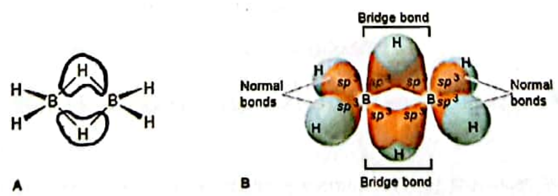
Forming Bridge Bonds with Electron-Poor Atoms

- In elemental boron and its many hydrides (boranes), there is no electron-rich atom to supply boron with electrons.

In these substances, boron attains an octet through some unusual bonding.

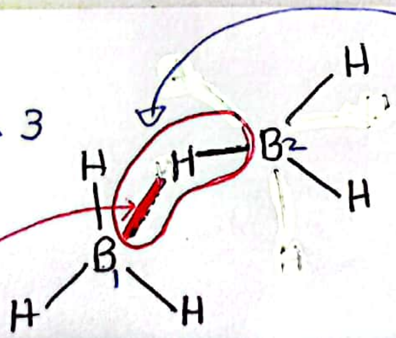
- In diborane (B_2H_6) and many larger boranes, for example, two types of B- H bonds exist.
  - The first type is a typical electron-pair bond.

banana
3 centre -3 electron bond



  - A sp^3 orbital of B overlapping a 1s orbital of H in each of the four terminal B-H bonds, using two of the three electrons in the valence level of each B atom.

${}_5B \longrightarrow 1s^2\underbrace{2s^22p^1}_{\text{valence electrons} \rhd 3}$



When B_1 forms a bond with 3 H atoms there are no electrons left without bonding

So in this bond there are only 2 electrons are envolved for bond forming.

Highlights of Boron Chemistry

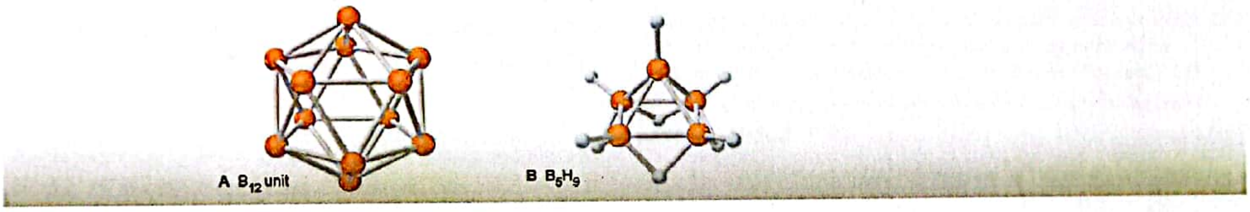
- Other type of bond is a hydride bridge bond (or three-center, two electron bond)

Each B-H-B grouping is held together by only two electrons. Two sp^3 orbitals, one from each B, overlap an H 1s orbital between them.

- Two electrons move through this extended bonding orbital—one from one of the B atoms and the other from the H atom—and join the two B atoms via the H atom bridge.

- Notice that each B atom is surrounded by eight electrons: four from the two normal B±H bonds and four from the two B-H-B bridge bonds.

- In many boranes and in elemental boron, one B atom bridges two others in a three-center, two-electron B-B-B bond.



Diagonal Relationships: Beryllium and Aluminum

- Beryllium in Group 2A(2) and aluminum in Group 3A(13) are another pair of diagonally related elements. Both form oxoanions in strong base:

  beryllate, $Be(OH)_4^{2-}$, and aluminate, $Al(OH)_4^{2-}$

- Both have bridge bonds in their hydrides and chlorides.

- Both form oxide coatings impervious (impermeable) to reaction with water, and both oxides are amphoteric, extremely hard, and high melting.

- Although the atomic and ionic sizes of these elements differ, the small, highly charged Be^{2+} and Al^{3+} ions polarize nearby electron clouds strongly. Therefore, some Al compounds and all Be compounds have significant covalent character.

Sodium Laureth Sulphate

$CH_3(CH_2)_{10}CH_2(OCH_2CH_2)_nOSO_3Na$

is used in toothpaste, shampoo and handwash

Some Reactions and Compounds

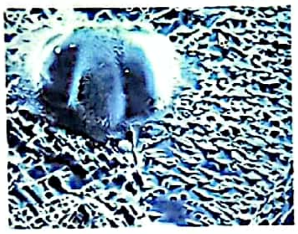
Important Reactions

In reactions 1 to 3, the elements (E) usually require higher temperatures to react than those of Groups 1A(1) and 2A(2); note the lower oxidation state of Tl. Reactions 4 to 6 show some compounds in key industrial processes.

1. The elements react sluggishly, if at all, with water:

$$2Ga(s) + 6H_2O(hot) \longrightarrow 2Ga^{3+}(aq) + 6OH^-(aq) + 3H_2(g)$$
$$2Tl(s) + 2H_2O(steam) \longrightarrow 2Tl^+(aq) + 2OH^-(aq) + H_2(g)$$

Al becomes covered with a layer of Al_2O_3 that prevents further reaction (*see photo*).



Electron micrograph of oxide coating on Al surface

2. When strongly heated in pure O_2, all members form oxides:

$$4E(s) + 3O_2(g) \xrightarrow{\Delta} 2E_2O_3(s) \qquad (E = B, Al, Ga, In)$$
$$4Tl(s) + O_2(g) \xrightarrow{\Delta} 2Tl_2O(s)$$

Oxide acidity decreases down the group: B_2O_3 (weakly acidic) > Al_2O_3 > Ga_2O_3 > In_2O_3 > Tl_2O (strongly basic), and the +1 oxide is more basic than the +3 oxide.

3. All members reduce halogens (X_2):

$$2E(s) + 3X_2 \longrightarrow 2EX_3 \qquad (E = B, Al, Ga, In)$$
$$2Tl(s) + X_2 \longrightarrow 2TlX(s)$$

The BX_3 compounds are volatile and covalent. Trihalides of Al, Ga, and In are (mostly) ionic solids but occur as covalent dimers in the gas phase; in this way, the 3A atom attains a filled outer level.

4. Acid treatment of Al_2O_3 is important in water purification:

$$Al_2O_3(s) + 3H_2SO_4(l) \longrightarrow Al_2(SO_4)_3(s) + 3H_2O(l)$$

In water, $Al_2(SO_4)_3$ and other agents form a colloid that aids in removing suspended particles (*see Chemical Connections, p. 541*).

5. The overall reaction in the production of aluminum metal is a redox process:

$$2Al_2O_3(s) + 3C(s) \longrightarrow 4Al(s) + 3CO_2(g)$$

This process is carried out electrochemically in the presence of cryolite (Na_3AlF_6), which lowers the melting point of the reactant mixture and takes part in the change (*Section 22.4*).

6. A displacement reaction produces gallium arsenide, GaAs (*margin note, p. 578*):

$$(CH_3)_3Ga(g) + AsH_3(g) \longrightarrow 3CH_4(g) + GaAs(s)$$

Important Compounds

1. Boron oxide, B_2O_3. Used in the production of borosilicate glass (*see margin note, p. 582*).

2. Borax, $Na_2[B_4O_5(OH)_4]\cdot 8H_2O$. Major mineral source of boron compounds and B_2O_3. Used as a fireproof insulation material and as a washing powder (20-Mule Team Borax).

3. Boric acid, H_3BO_3 [or $B(OH)_3$]. Used as external disinfectant, eyewash, and insecticide.

4. Diborane, B_2H_6. A powerful reducing agent for possible use as a rocket fuel. Used to synthesize higher boranes, compounds that led to new theories of chemical bonding.

5. Aluminum sulfate (cake alum), $Al_2(SO_4)_3$. Used in purifying water, tanning leather, and sizing paper; as a fixative for dyeing cloth (*see photo*); and as an antiperspirant.

6. Aluminum oxide, Al_2O_3. Major compound in natural source (bauxite) of Al metal. Used as abrasive in sandpaper, sanding and cutting tools, and toothpaste. Large crystals with metal ion impurities often of gemstone quality (*see photo*). Inert support for chromatography. In fibrous forms, woven into heat-resistant fabrics; also used to strengthen ceramics and metals.

Ruby

7. $Tl_2Ba_2Ca_2Cu_3O_{10}$. Becomes a high-temperature superconductor at 125 K, which is readily attained with liquid N_2 (*see photo*).

Magnet levitated by superconductor